



AI XAU Trader Pro

An autonomous, precision-first trading system for Gold (XAUUSD) — engineered for statistical certainty, capital preservation, and full broker parity.

AI-POWERED

XAUUSD

METATRADER 5

The Problem with Human Trading

Emotion-Driven Errors

Fear and greed override logic — the #1 cause of retail trading losses

Execution Latency

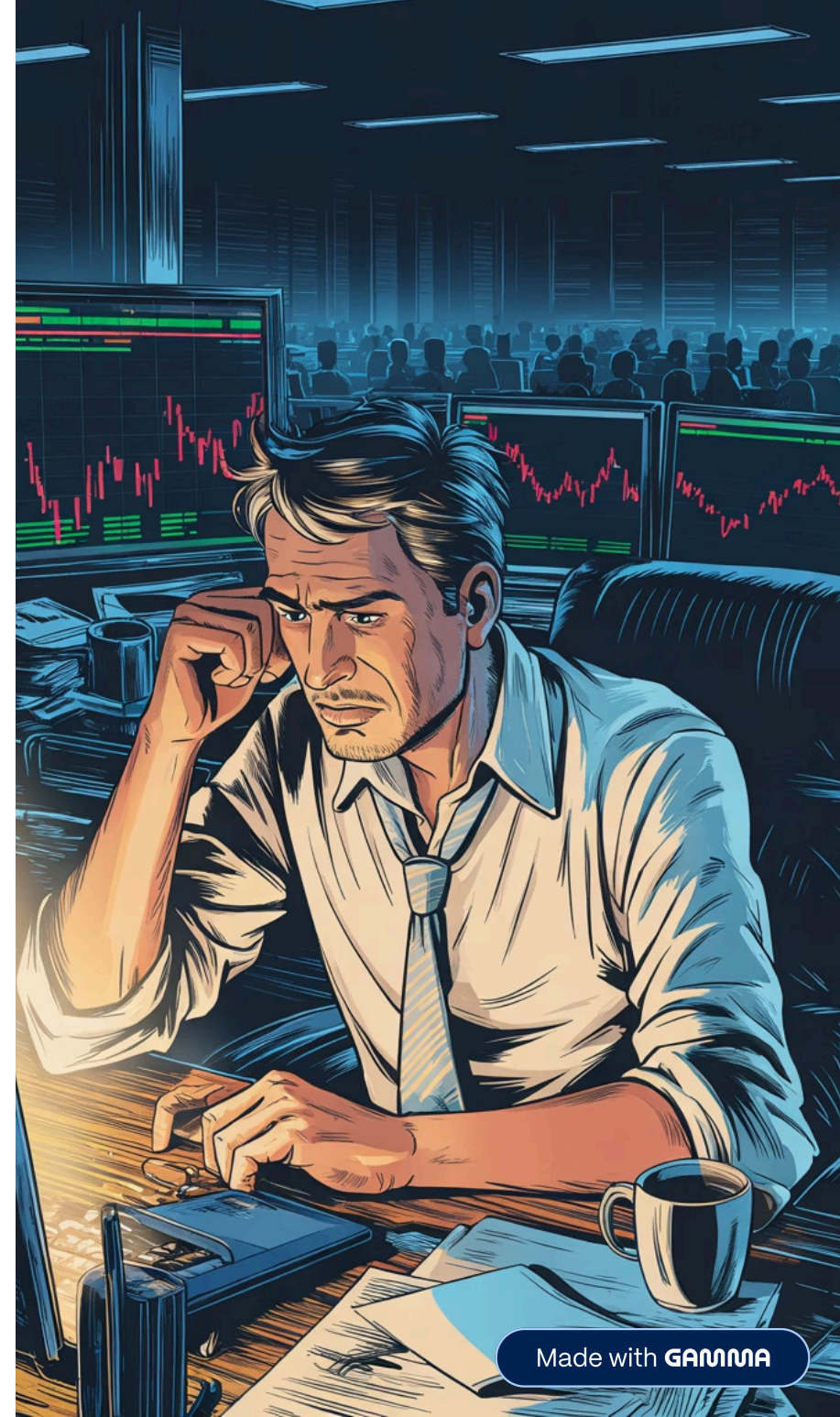
Manual entries miss optimal price by seconds, compressing edge

Inconsistent Risk

Ad-hoc lot sizing and ignored stop-losses destroy accounts during volatility

No Systematic Edge

Gut-feel decisions lack the statistical rigor required for long-term profitability



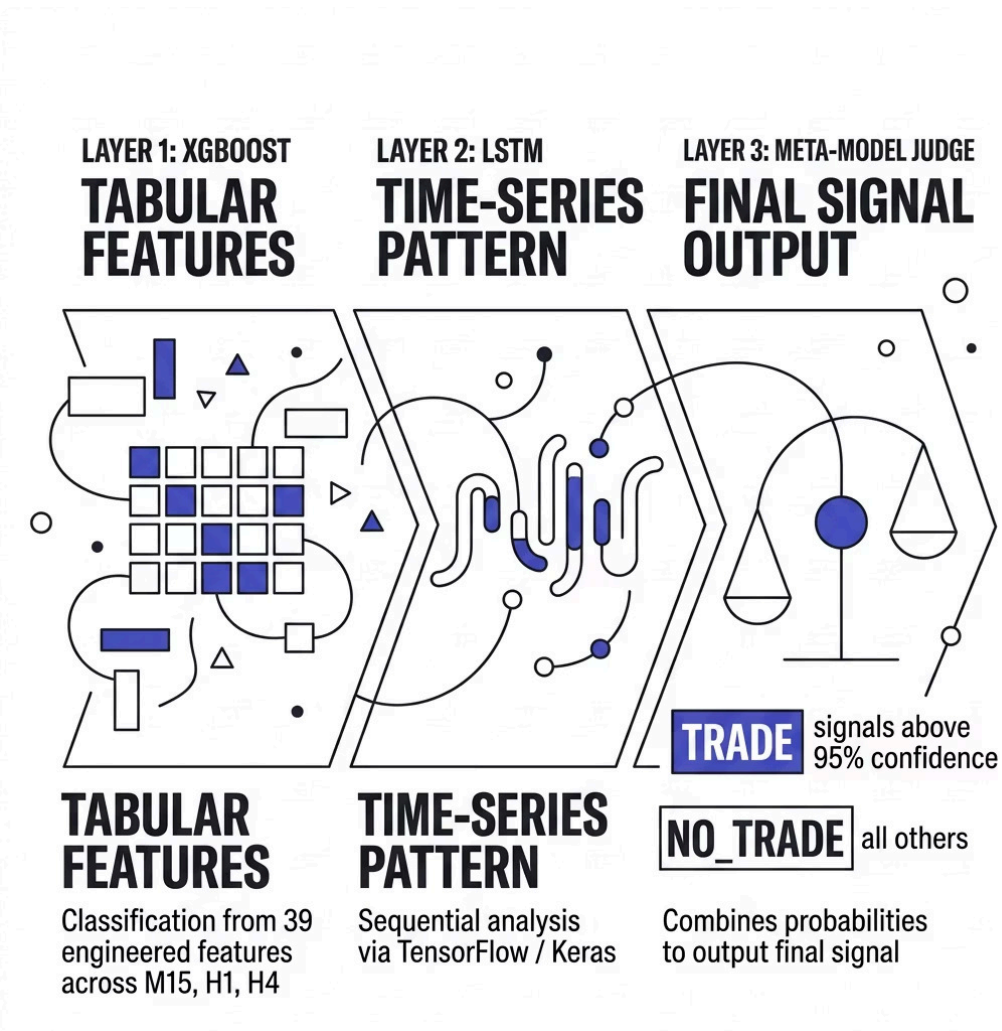
System Architecture

A fully integrated, three-tier stack where every layer speaks the same truth — the MT5 broker.



All balance, equity, tick, and deal history data is pulled directly from MT5 — eliminating reconciliation errors and ensuring a single source of truth across the entire system.

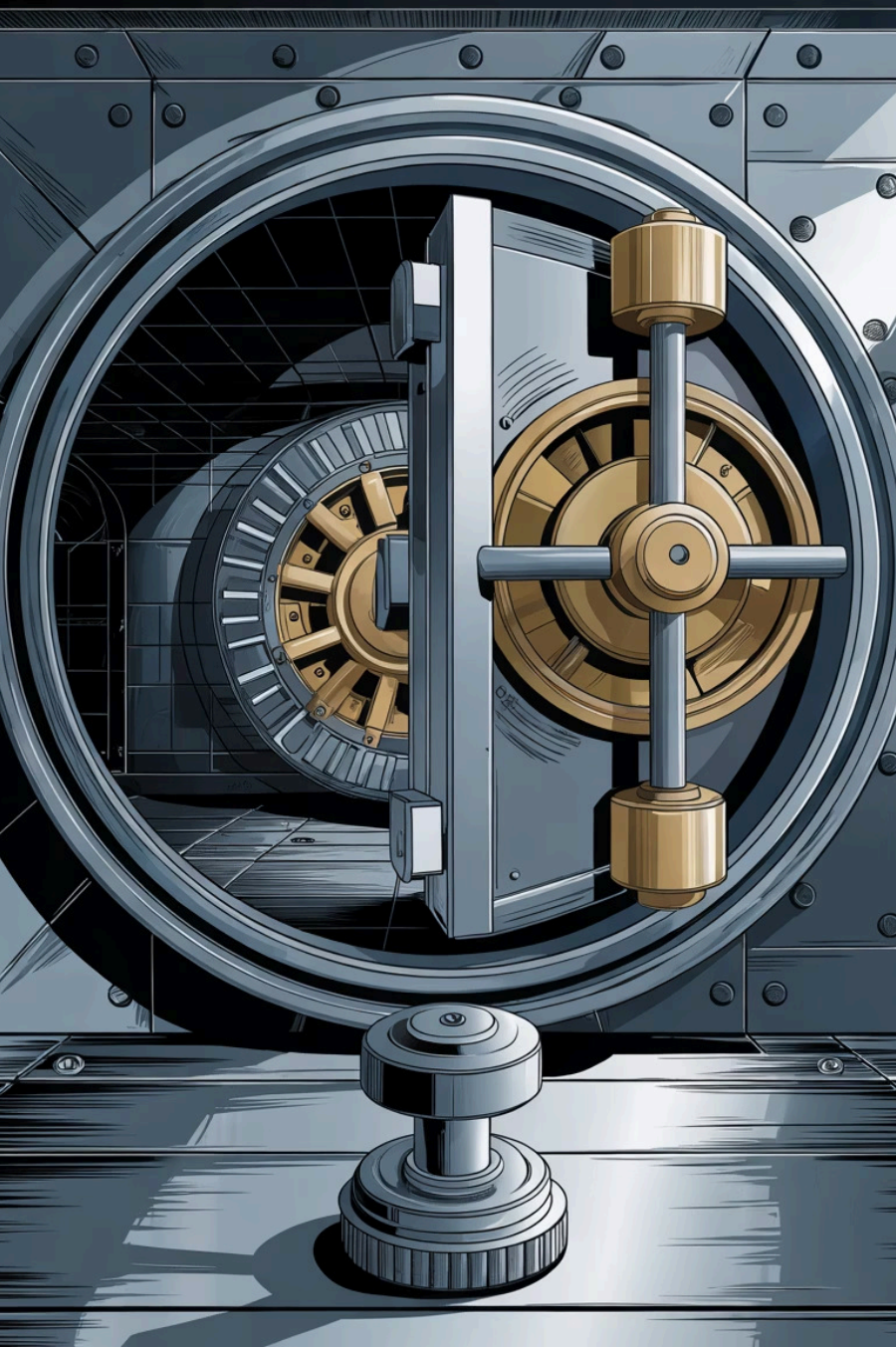
The AI Brain: 39-Feature Ensemble



Multi-Timeframe Inputs

- M15, H1, H4 data ingested simultaneously
- RSI, EMA50/200, EMA slopes, ATR
- Dynamic Support & Resistance levels
- Moving average distance features

⚠ 95% Confidence Gate — Anything below threshold returns a hard **NO_TRADE** signal. The bot never guesses.



Precision Over Frequency

Act only when the edge is undeniable.

The Meta-Model's 95% confidence threshold means the bot sits idle rather than force suboptimal trades — a discipline most human traders cannot sustain.

Risk Management Framework



Fixed TP / SL

5.0 Take Profit · 3.0 Stop Loss — hardcoded R:R ratio, never overridden



Dynamic Lot Sizing

0.01 lots per \$100 of account balance — auto-compounds as the account grows



Institutional Cooldown

3 consecutive losses trigger a mandatory 1-hour trading pause — prevents algorithmic spiraling



Liquidity Window

Trades restricted to 08:00–18:00 UTC — peak institutional volume, tighter spreads

Real-Time Dashboard



Built for Clarity Under Pressure

- **100% MT5 Parity** — Trade history mirrors broker terminal exactly, last 10 closed positions
- **Live WebSockets** — Real-time candles, price ticks, balance & equity, zero page reloads
- **AI Status Pipeline** — Visual thought process: Collecting → Analyzing → Running Ensemble → Confidence Generated
- **Cooldown Timer** — Full-screen countdown replaces pipeline display during safety pause

 Premium dark-mode aesthetic with glassmorphism and dynamic micro-animations.

AI Status Pipeline: Live Transparency

Analyzing Momentum

Compute RSI, EMAs, ATR,
S/R levels.

Running Ensemble

XGBoost + LSTM combined
into meta-model.

Collecting Data

Pull M15/H1/H4 candles and
ticks.

Confidence Generated

$\geq 95\%$ execute trade;
otherwise NO_TRADE.

Every scan cycle is made visible to the operator — no black-box anxiety. When the Institutional Cooldown activates, the pipeline is replaced by a high-visibility digital countdown timer showing exactly when scanning resumes.

Technology Stack



Python 3.11 + FastAPI

Async routing, background task execution, WebSocket server



XGBoost + TensorFlow

Ensemble model stack with LSTM for sequential pattern recognition



MetaTrader 5 API

Official Python package — direct broker integration, live data pull



Vanilla HTML/JS/CSS

Zero-dependency frontend — fast, lightweight, premium dark UI

What's Next

→ Live Deployment

Connect to a live MT5 account with real capital under supervised oversight

→ Model Retraining Pipeline

Automated weekly retraining on new XAUUSD data to maintain edge as market regimes shift

→ Multi-Asset Expansion

Extend the ensemble architecture to EURUSD, crude oil, and equity indices

→ Portfolio-Level Risk Engine

Aggregate drawdown controls and cross-asset correlation monitoring across all deployed bots

